

MISHELINA JOBY

LinkedIn

DOB – 18/09/2005

mishelinajoby.mec@gmail.com

+91 7736084864

SKILLS & INTERESTS

- **Technical Skills:** Verilog, Arduino, Vivado, LTSpice, Proteus, Embedded Systems, Scripting Languages (Python), C++, HTML
- **Soft Skills:** Communication, Problem Solving, Teamwork, Critical Thinking, Attention to Detail, Leadership, Adaptability, Time Management
- **Interests:** FPGA Programming (Verilog), Signal and Image Processing, Embedded Systems and IoT

EDUCATION

- **Govt. Model Engineering College** **9.27 | 2027**
KTU, B.Tech in Electronics and Communication Engineering
- **Mar Kauma Higher Secondary School** **99.3% | 2023**
State, 12th
- **Vimala Central School** **98% | 2021**
ICSE, 10th

PROJECTS

- **Traffic Light Controller** **Team Size: 3**
Team Member **4 Days**
Technologies Used: Cadence tools, Verilog, Vivado
Designed and developed a 4-way Traffic Light Controller using FPGA to manage traffic signal sequences, implementing a Finite State Machine (FSM) and sequential logic for precise timing and state transitions.
- **Digital Stopwatch Using FPGA** **Team Size: 2**
Team Lead **2 Weeks**
Technologies Used: Verilog, Vivado, FPGA
Designed and developed a Digital Stopwatch Using FPGA to perform start, stop and reset operations with counting capability up to 99 seconds, implementing sequential logic and FPGA-based control.
- **RC Plane Project** **Team Size: 9**
Team Member **3 Months**
Technologies Used: Aerodynamics, Electronic Control Systems, Radio Communication
Collaborated on the development of the RC Plane Project to design and assemble a radio-controlled aircraft, integrating components such as a DC motor, propeller, receiver, transmitter and control surfaces while applying principles of aerodynamics and electronic flight control.
- **Civic Issue Reporting Application** **Team Size: 4**
Team Member **2 Weeks**
Technologies Used: Flutter, Dart, Firebase, Google Maps API
Designed and developed a Civic Issue Reporting Application using Flutter and Firebase to enable citizens to report issues such as potholes, waste dumping and streetlight failures with photo-based reporting and automatic GPS location detection.
- **ECG Noise Cancellation Using Verilog** **Team Size: 4**
Team Member **12 Weeks**
Technologies Used: Verilog, Vivado, Python
Developed a Digital Signal Processing sequence to filter baseline drift and muscle interference from raw ECG signals using a combination of high-pass and low-pass filters.

COURSES AND CERTIFICATIONS

- Pursuing a **Minors Degree in Computer Science and Engineering** from **KTU**.
- Completed **Rapid Prototyping Using FPGA Training** conducted by **Indian Institute of Information Technology Kottayam**, gaining hands-on exposure to FPGA development.
- Successfully completed the **NPTEL** certified course **Embedded System Design With ARM**, conducted by **IIT Kharagpur**, covering ARM architecture, embedded programming and system-level design concepts.
- Obtained certification in **VLSI Design Flow: RTL To GDS** from **NPTEL**, offered by **IIT Madras**, demonstrating proficiency in RTL design, synthesis, verification and physical design stages of the VLSI design flow.

ACHIEVEMENTS AND ACTIVITIES

- Attended a **Workshop on PCB Design**, conducted by **Cochin University of Science and Technology**.
- Attended a **Workshop on VLSI DESIGN FLOW: RTL TO GDSII USING CADENCE TOOLS**, conducted by **IEEE CASS KERALA CHAPTER**.
- **Hobbies:** Stitching

REFERENCES

- **Prof. Dr. Jayachandran E S**, Principal, Govt. Model Engineering College, Kochi, Email ID: principal@mec.ac.in
- **Prof. Dr. Jobymol Jacob**, HOD, Electronics and Communication Engineering, Govt. Model Engineering College, Kochi, Email ID: hodec@mec.ac.in